

Vinayak Manoj Nair

Aerospace Engineering · GNC / State Estimation · Flight Software V&V · Structural Analysis

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01 Summary

Aerospace Engineering student at Texas A&M University (B.S. May 2027, Fast Track BS/MS) working across guidance/navigation/control, flight-software verification, structural analysis and systems engineering. Builds projects around executable evidence: deterministic simulation, hardware test, fault injection, Monte Carlo, requirements traceability and explicit claim boundaries.

02 Education

Texas A&M University, B.S. Aerospace Engineering

College Station, TX · Expected May 2027 · GPA 3.45 / 4.00

Texas A&M University, M.S. Aerospace Engineering (Fast Track BS/MS)

In progress · Focus: GNC and spacecraft systems

03 Experience

Spacecraft Systems Engineer, TAMU-SPIRIT Capstone

Aegis Aerospace / ISS Payload Program · College Station, TX · Jan 2026 to Present

- Maintained 100% milestone compliance for 4 ISS-bound experiments targeting a Sep 2028 SpaceX Dragon launch by managing engineering change requests and NASA compliance tracking through SCR, SRR and PDR design reviews.
- Reduced thermal design risk for the DASH payload with material-selection trades and thermal analysis across -120°C to $+120^{\circ}\text{C}$ LEO cycling.
- Coordinated quality/configuration documentation across an 8-person engineering team as the program moved toward hardware manufacturing.

Controls & Embedded Systems Engineer

BLNC Turtle Robotics · College Station, TX · Aug 2025 to Dec 2025

- Reduced motor-validation errors by standardizing LabVIEW bench-test procedures and maintaining as-built documentation.
- Built and autonomously stabilized a two-wheeled inverted-pendulum rover using cascaded PID control.

Undergraduate Research Assistant

Texas A&M ACE Lab · College Station, TX · Dec 2023 to Dec 2024

- Cut manual analysis time by 40% and accelerated report delivery by 2 days with a Python pipeline processing 5,000+ data points per run.
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04 Selected engineering projects

GHOST-X: Occlusion-Tolerant Target State Estimation

ROS 2 · Raspberry Pi · C++ / Python · IMM · deterministic GNC SIL

- Developed a target-estimation and evidence pipeline centered on a formal IMM estimator with a bounded GHOST-MH comparison architecture, exposing covariance / dropout state through real visual measurement loss.
- Integrated Raspberry Pi + AprilTag hardware evidence with deterministic controlled-truth evaluation and closed-loop TRACKING / PREDICTION / SAFE_HOLD software-in-the-loop scenarios; hardware and software-truth accuracy are reported separately.

ASTRA-OS: Spacecraft-Style Flight Software Assurance

C++17 · Python · CMake / CTest · UDP command/telemetry · FDIR

- Built deterministic modes, command integrity/freshness guards, watchdog/health monitoring, FDIR, bounded recovery, binary telemetry and Python ground tooling.
- Current managed assurance baseline passes 20/20 native CTest suites, 8/8 deterministic scenarios, 10/10 FDIR cases, 24/24 protocol-conformance cases and 25/25 seeded Monte Carlo regressions; preserved Raspberry Pi execution evidence is explicitly separated from real-time qualification.

AeroFrame-DT: Aircraft Fitting Stress Substantiation

Hand analysis · FE verification · MMPDS · fatigue / damage tolerance · configuration control

- Carried a synthetic, non-OEM aircraft fitting from load basis through classical margins, FE verification, allowables governance, fatigue/damage tolerance, manufacturing and a revision-aware digital thread; all 18 formal requirements verified.
- Governing baseline margin is +0.151 after the analysis exposed invalid assumptions and forced a material reselection to 7050-T7451; retained A-basis-consistent method-scatter case is negative at -0.037.

F-16-Inspired Flight Controls Simulation

Python · fixed-wing dynamics · autopilot · bounded actuators · uncertainty analysis

- Built an educational software-in-the-loop flight-dynamics/autopilot framework with seeded sensor-error injection, gusts, actuator limits and a 50-run uncertainty campaign.
- Explicitly scoped as representative controls-software verification practice, not a validated F-16 model or aircraft-performance claim.

05 Technical skills

PROGRAMMING	C++17, Python, MATLAB, LabVIEW, Bash, CMake
GNC / ESTIMATION	Kalman filtering, IMM estimation, flight dynamics, PID/cascaded control, Monte Carlo, sensor fusion, closed-loop SIL
FLIGHT SOFTWARE / TEST	CTest, binary command/telemetry, FDIR, watchdogs, fault injection, requirements traceability, sanitizers/static analysis
STRUCTURES	Classical stress methods, FEA verification, fatigue, damage tolerance, modal/buckling, tolerances, allowables governance
SYSTEMS / TOOLS	Requirements, verification planning, design reviews, configuration/change control, Linux, Git, Raspberry Pi, ROS 2